Smart Work Beats Hard Work

APTITUDE SHORTCUTS
CHEAT SHEET

Think
Practice
Crack!

① PERCENTAGE (%)

★ % of y = $\frac{x}{100} \times y$

★ Increase % = $\left(\frac{New - Old}{Old}\right) \times 100$

★ Decrease % = $\left(\frac{Old - New}{Old}\right) \times 100$

★ Successive Change = $\frac{(100+a)(100-b) - 100}{100}$

TIPS

- 10% of a number → move 1 step left (÷10)
- 1% of a number → move 2 step left (÷100)
- 5% of a number → half of 10%

② PROFIT & LOSS

★ Profit % = $\left(\frac{SP - CP}{CP}\right) \times 100$

★ Loss % = $\left(\frac{CP - SP}{CP}\right) \times 100$

MARKED PRICE (MP)

SP = MP × $\left(\frac{100 - Discount\%}{100}\right)$

TRICKS

- 10% Profit ⇒ SP = $\frac{11}{10} \times CP$
- 10% Loss ⇒ SP = $\frac{9}{10} \times CP$
- 25% Profit ⇒ SP = $\frac{5}{4} \times CP$
- 25% Loss ⇒ SP = $\frac{3}{4} \times CP$

③ TIME & WORK

★ Work = Time × Efficiency

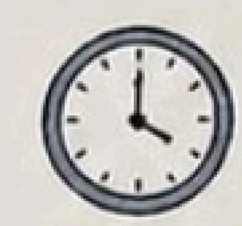
★ If A alone takes x days,
A's 1 day work = $\frac{1}{x}$

★ A + B together time = $\frac{1}{x} + \frac{1}{y} = \frac{xy}{x+y}$

★ If A is n times more efficient than B
⇒ Time taken by A alone = $\frac{\text{Time by B alone}}{n}$

TRICKS

A alone 1 day = $\frac{1}{x}$
A alone 1 hour = $\frac{1}{24x}$
A alone 1 min = $\frac{1}{1440x}$



Time is Money

④ SPEED, DISTANCE & TIME

- Speed = $\frac{Distance}{Time}$
- Time = $\frac{Distance}{Speed}$
- Distance = Speed × Time

TRICKS

- If speed is increased by x%
⇒ Time is decreased by $\frac{x}{100+x}$ %
- If speed is decreased by x%
⇒ Time is increased by $\frac{x}{100-x}$ %
- Average Speed = $\frac{Total\ Distance}{Total\ Time}$

⑤ NUMBER SYSTEM

★ Even numbers → 0, 2, 4, 6, 8
★ Odd numbers → 1, 3, 5, 7, 9
★ Divisibility Rules (Quick)

2 → Last digit even
3 → Sum of digits divisible by 3
4 → Last two digits divisible by 4
5 → Last digit 0 or 5
6 → Divisible by 2 and 3
8 → Last three digits divisible by 8
9 → Sum of digits divisible by 9
10 → Last digit 0
11 → Add digits, put sum in middle (carry if needed)

⑥ PROBABILITY

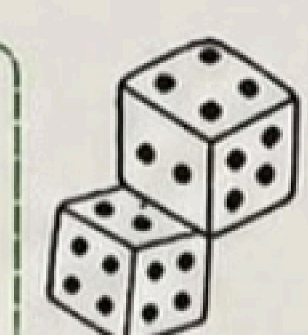
★ P(E) = $\frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$

★ $P(\bar{E}) = 1 - P(E)$

★ If events are independent,
P(A and B) = P(A) × P(B)

TRICKS

- Coin Toss : P(Head) = $\frac{1}{2}$
- Two Coins : P(2 Heads) = $\frac{1}{4}$
- Two Dice : Total outcomes = 36



⑦ GEOMETRY FORMULAS

□ Square – Area = a^2, Perimeter = 4a
▭ Rectangle – Area = $\ell \times b$, Perimeter = $2(\ell+b)$
○ Circle – Area = πr^2, Circumference = $2\pi r$
△ Triangle – Area = $\frac{1}{2} \times b \times h$
◺ Right Triangle – $H^2 = P^2 + B^2$

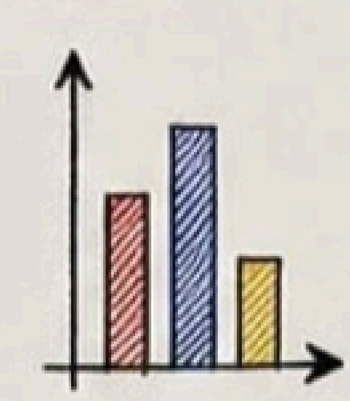
$\pi = \frac{22}{7}$ (Use 3.14 for easy calc.)

⑧ IMPORTANT SQUARES & CUBES

SQUARES (1 to 20)		CUBES (1 to 10)
1^2 = 1	11^2 = 121	1^3 = 1
2^2 = 4	12^2 = 144	2^3 = 8
3^2 = 9	13^2 = 169	3^3 = 27
4^2 = 16	14^2 = 196	4^3 = 64
5^2 = 25	15^2 = 225	5^3 = 125
6^2 = 36	16^2 = 256	6^3 = 216
7^2 = 49	17^2 = 289	7^3 = 343
8^2 = 64	18^2 = 324	8^3 = 512
9^2 = 81	19^2 = 361	9^3 = 729
10^2 = 100	20^2 = 400	10^3 = 1000

⑨ DI (DATA INTERPRETATION)

- First read Questions
- Look for Units in chart
- Use Approximation
- Compare in % (not in value)
- Use Elimination Method
- Don't calculate exact, calculate smart



HANDY TIPS

- 1 Crore = 100 Lakhs
- 1 Lakh = 100,000
- 1 Thousand = 1,000
- 1 Million = 10 Lakh

10. EXAM STRATEGY (TCS & INFOSYS)

✓ Manage Time → 1 Ques < 1 Minute
✓ Attempt Easy First
✓ Don't get stuck – Move Ahead
✓ Use Elimination
✓ Accuracy > Speed
✓ Revise if time remains

Shortcuts Save Time
Practice Makes Perfect

IMPORTANT REMINDERS

- ☐ Formulae yaad rakho
- ☐ Tables (2 to 20) pakka karo
- ☐ Daily 30 min practice
- ☐ Analyze your mistakes
- ☐ Stay Calm & Confident

ALL THE ☆ BEST! ☆

YOU GOT THIS!

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Keep Practicing, Keep Growing!